1 (a) (i) metre rule / tape (*not 'rule'*) B1 [1]

(ii) micrometer (screw gauge) / digital caliper B1 [1]

(iii) ammeter and voltmeter / ohmmeter / multimeter on 'ohm' setting B1 [1]

(b) (i) resistivity $= RA / L$ C1
$= [7.5 \times \pi \times (0.38 \times 10^{-3})^2 / 4] / 1.75$ M1
$= 4.86 \times 10^{-7}\ \Omega\ \mathrm{m}$ A0 [2]

(ii) (uncertainty in R =) $[0.2 / 7.5] \times 100 = 2.7\%$
<u>and</u> (uncertainty in L =) $[3 / 1750] \times 100 = 0.17\%$ C1
(uncertainty in A =) $2 \times (0.01 / 0.38) \times 100 = 5.3\ \%$ C1
total = 8.13% C1

uncertainty $= 0.395 \times 10^{-7}$ (Ω m) A1 [4]
(*missing 2 factor in uncertainty in A, then allow max 3/4*)

(c) resistivity $= (4.9 \times 10^{-7} \pm 0.4 \times 10^{-7})\ \Omega\ \mathrm{m}$ A1 [1]

2 (a) work done is the force × the distance moved / displacement in the direction of the